RDS R Packages

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Introduction

The RDS team presents a suite of R packages tailored for epidemiological data management, analysis, and visualisation. These packages address the complex challenges inherent in epidemiological surveillance and incident support.

Within our suite, epidm and epiviz play distinct yet complementary roles. epidm focuses on managing and transforming epidemiological data, offering standardised methods for processing key datasets used by UKHSA. Meanwhile, epiviz specialises in data visualisation, providing a rich set of functions to create insightful representations of epidemiological trends and patterns.

Aligned with our vision, these packages aim to establish a cohesive ecosystem that empowers UKHSA teams with the necessary tools and insights for informed decision-making. With a focus on standardisation, interoperability, and user-friendliness, our packages highlights our commitment to advancing data-driven decision-making and public health outcomes.

Key Objectives

1. Standardisation: We aim to establish standard methodologies and practices for handling epidemiological

data, ensuring consistency and reliability across analyses and projects.

2. Interoperability: Our packages are designed to seamlessly integrate with each other and with existing tools and datasets, fostering collaboration and knowledge sharing within the organisation.
3. Accessibility: We strive to make our packages accessible to a organisation wide through external repositories such as CRAN.
4. Innovation: We are committed to continuous innovation, refining existing functionalities, and developing new features that address emerging challenges and opportunities in epidemiological data analysis and visualisation.
5. Community Engagement: We seek to actively engage with the R community in UKHSA, soliciting feedback and collaborating on development efforts.

Code Reusability and Maintenance Strategy

Both epidm and epiviz are developed with a focus on code reusability and maintainability. Shared functions and modular design principles are employed to maximise code reuse across projects and minimise duplication of efforts. Regular code reviews, documentation updates, and version control practices ensure the reproducibility and long-term sustainability of our reproducible analytical pipelines.

epidm: Epidemiological Data Management

The epidm package aims to provide standardised methods for the management and transformation of epidemiological data, particularly focusing on dataset held by UKHSA.

epidm facilitates the processing of key epidemiological datasets, including infection data from the Second Generation Surveillance System (SGSS), hospital data such as Emergency Care Data Set (ECDS), Secondary Use

Services (SUS), and Hospital Episode Statistics (HES). By offering functions for cleaning, processing, and linking these datasets, epidm streamlines the data preparation phase for subsequent analysis and surveillance efforts.

Uses of epidm:

- Hospital data linkage for MPOX, STEC incidents
- Data linkage and de-duplication of datasets in RDS Shiny dashboards

epidm is currently available through CRAN: https://cran.r-project.org/web/packages/epidm/index.html

epiviz: Epidemiological Data Visualisation

epiviz aims to provide easy-to-use data visualisation functions for R data science products, facilitating both static and interactive visualisations.

epiviz offers a suite of visualisation functions tailored to epidemiological data, including line charts, column charts, epidemiological maps, and age-sex pyramids. Additionally, it provides helper functions to streamline the conversion of line lists into aggregated data frames, ready for visualisation.

Uses of epiviz:

- Annual TB regional report
- IID weekly bulletin

epiviz is currently being prepared for submission to CRAN: https://github.com/harshanal/epiviz

Product roadmap

epidm

Function	Description	REQ	IMP	DOC	TEST	EXP	GitHub?	CRAN?
Utility functions: Functions for reading or writing data from various sources								
sql_clean()	Clean and Read a SQL query	✅	✅	✅	✅	✅	✅	✅
sql_connect()	Connect to a SQL database	✅	✅	✅	✅	✅	✅	✅
sql_read()	Read a table from a SQL database	✅	✅	✅	✅	✅	✅	✅
sql_write()	Write a table to a SQL database	✅	✅	✅	✅	✅	✅	✅
csv_from_zip()	Download a csv from a zip	✅	✅	✅	✅	✅	✅	✅
lookup_recode()	Lookup table	✅	✅	✅	✅	✅	✅	✅

	switch handler							
Data Management Functions: Functions for handling, cleaning and processing public health data								
valid_nhs()	NHS Number Validity Check	✅	✅	✅	✅	✅	✅	✅
uk_patient_id()	Patient ID record grouping	✅	✅	✅	✅	✅	✅	✅
proxy_episode_dates()	HES/SUS Episode Date Cleaning	✅	✅	✅	✅	✅	✅	✅
cip_spells()	Continuous Inpatient (CIP) Spells	✅	✅	✅	✅	✅	✅	✅
group_time()	Grouping of intervals or events in time together	✅	✅	✅	✅	✅	✅	✅

respeciate_generic()	Respeciate unspecified samples	✅	✅	✅	✅	✅	✅	✅
inpatient_codes()	Inpatient Codes cleanup	✅	✅	✅	✅	✅	✅	✅
link_ae_inpatient()	Link A&E to Inpatient records	✅	✅	✅	✅	✅	✅	✅
Mortality data processing	Import and clean ONS mortality data from HPDW							
Lookup tables: **Lookup tables used to reclassify, regroup, or translate data**								
genus_gram_stain	Bacterial Genus Gram Stain Lookup Table	✅	✅	✅	✅	✅	✅	✅

respeciate_organism	Respeciated organisms	✅	✅	✅	✅	✅	✅	✅
specimen_type_grouping	Specimen type grouping	✅	✅	✅	✅	✅	✅	✅

Key

REQ: Requirement capture, IMP: Function implementation, DOC: Function documentation, TEST: Unit test implementation, EXP: Examples documented, GitHub: Available to use through GitHub Version, CRAN?-Available to use through CRAN version

epiviz

Function	Description	REQ	IMP	DOC	TEST	EXPERIMENTAL	GitHub?	CRAN?
Data visualisation functions								
agesex_pyramid()	Age Sex Pyramid Dynamic and Static	✅	✅ only static	✅	✅		✅	✅

col_chart()	Column Chart Dynamic and Static	✅	✅	✅	✅	✅	✅	
epi_map()	epi_map Dynamic and Static	✅	✅	✅	✅		✅	✅
epi_curve()	epi_curve Dynamic and Static	✅	✅	✅	✅		✅	✅
line_chart()	Line_chart Dynamic and Static	✅	✅	✅	✅		✅	✅
point_chart()	Point chart Dynamic and Static	✅	✅	✅	✅		✅	
llm_intepret()	Generate narrative for	✅	✅	✅	✅	✅	✅	

	data frame or ggplot object							
Example datasets								
lab_data	Synthetic Lab Data for EpiViz functions	✅	✅	✅	✅		✅	✅

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